

(TR-104) PYTHON WITH AI/ML

Training Day 87 Report

Date: 12 May 2026

Topic Covered: Authentication Module Development, Streamlit Integration, AI System Testing & Frontend Validation

The eighty-seventh day of training focused on developing the authentication module for the college project, integrating frontend and backend components, enhancing user authentication workflows, and performing comprehensive testing of the AI software system.

- Designed and implemented complete login and registration workflows for the college project.
- Developed session-based authentication management using Streamlit session state.
- Integrated frontend authentication screens with backend database functions for seamless user management.
- Created reusable backend functions for account creation, login validation, and username verification.
- Implemented secure password hashing and verification mechanisms using bcrypt.
- Configured database interaction workflows for authentication and user data management.
- Developed multi-screen navigation between login, registration, home screen, and dashboard interfaces.
- Implemented conditional rendering logic based on authentication state and user sessions.
- Added validation for empty fields, duplicate usernames, invalid credentials, and password mismatch scenarios.
- Enhanced user experience through success messages, error prompts, toast notifications, and dynamic UI updates.
- Improved frontend structure and organized authentication-related components for better maintainability.

- Modularized backend and frontend authentication logic to support scalable development.
- Reviewed and optimized frontend-backend communication flow for authentication-related operations.
- Performed testing of the MB AI software frontend and validated AI response rendering behavior.
- Tested customer, trailer, payment, and analytical queries to verify response accuracy and frontend integration.
- Conducted prompt injection testing, reverse prompt testing, and irrelevant query validation.
- Verified fallback handling for out-of-context queries and analyzed system behavior across multiple query scenarios.
- Tested frontend-backend communication reliability and response consistency for AI-driven interactions.
- Revised concepts related to authentication workflows, Streamlit session management, bcrypt password security, frontend-backend integration, AI system testing, and prompt validation techniques.

Overall, the training enhanced understanding of authentication system development, secure user management, Streamlit-based application workflows, AI software testing, and frontend-backend integration strategies.